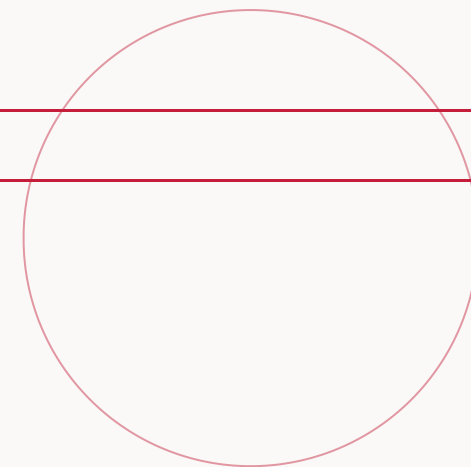


Correlation, Regression, Beta & Alpha



*From a pair of stocks
to the market model.*

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A Course on Quantitative Finance

MOTIVATION

A stock never lives alone

So far we measured each stock's *return* and *risk* in isolation. But every stock is buffeted by the same market — some news lifts every boat, some lowers it. We need tools that read those *joint* movements.

- How tightly do **two stocks** move together?
- How much of a stock's risk is **the market's risk**?
- What return is left after we **strip out the market**?
- Are those leftovers **luck**, or skill?

Three weekly return series, one calendar

One year of **Friday-close weekly returns** for two stocks and the market — aligned on the same dates.

STOCK 1

SBI

$n = 51$

State Bank of India

STOCK 2

Yes Bank

$n = 51$

Yes Bank Ltd.

MARKET

Sensex

$n = 51$

BSE benchmark index

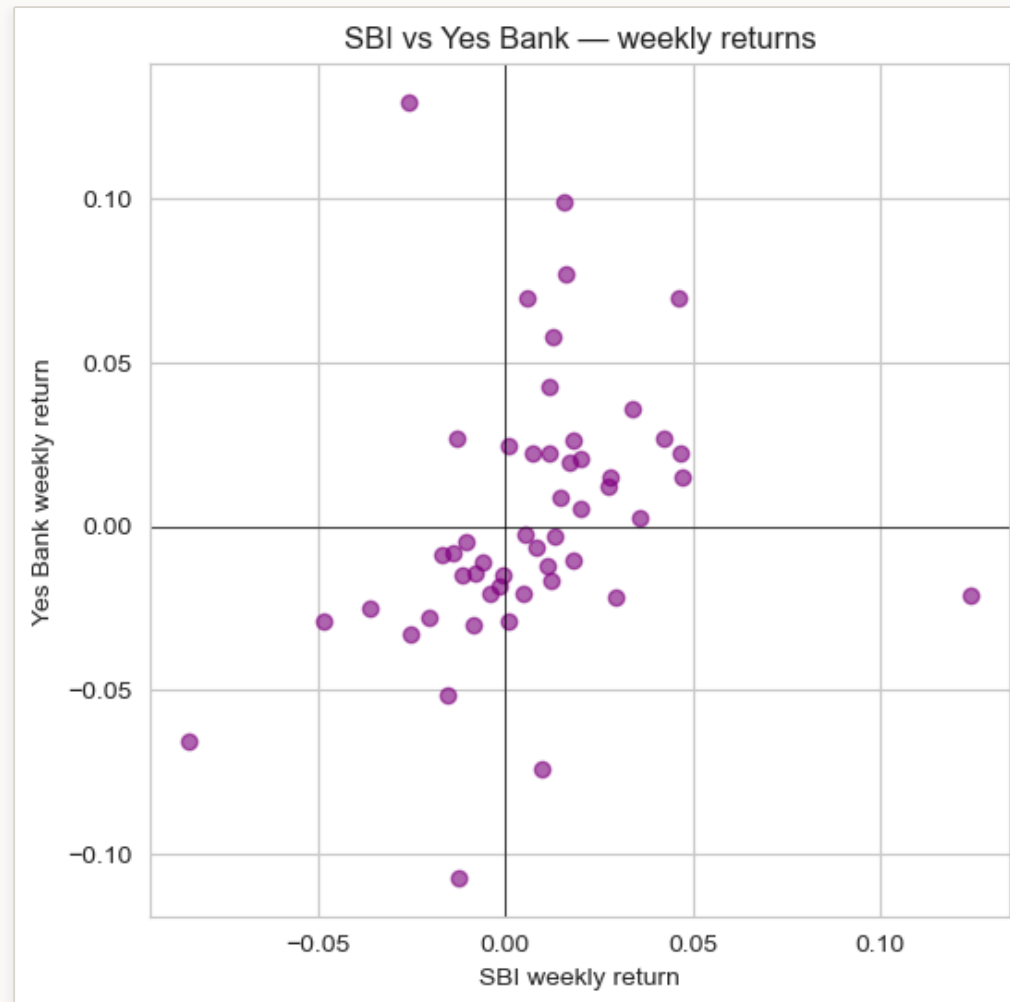
Pairs first — then each stock against the market.

CHAPTER I

Correlation — *how tightly do they* *move together?*

SBI vs Yes Bank — weekly returns

- ▶ Cloud **tilts upward** — loose positive link
- ▶ A lot of **scatter** around the trend
- ▶ Eyeball estimate of ρ : ~ 0.3



DEFINITION

Pearson correlation

Symmetric, unit-free, and bounded on $[-1, +1]$. It measures the strength of **linear** association — no direction of cause, no units, no scale.

POPULATION & SAMPLE CORRELATION

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad \hat{\rho} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Divide the covariance by both standard deviations — that is the only thing that strips the units.

+1 = move in lockstep · **0** = no linear link · **-1** = perfect mirror.

RESULTS · PAIRWISE P

Three pairs, three numbers

	SBI	YES BANK	SENSEX
<i>SBI</i>	1.000	0.285	0.481
<i>Yes Bank</i>	0.285	1.000	0.479
<i>Sensex</i>	0.481	0.479	1.000

*Each stock is **more correlated with the market** (~0.48) than with the other stock (~0.29). The market is the shared driver — once you remove it, the residual link between the two stocks is weak.*

A BRIDGING IDENTITY

$$R^2 = \rho^2$$

Correlation and the *fit quality* of a single-predictor regression are **two views of the same thing**.

THE IDENTITY

$$R^2 = \rho_{X,Y}^2 = \text{slope}(Y \sim X) \cdot \text{slope}(X \sim Y)$$

Square the correlation and you get the share of variance the regression explains. The product of the two opposite-direction slopes is also ρ^2 .

$\rho \approx 0.48$ between SBI and Sensex implies $R^2 \approx 0.23$ — the market explains roughly a quarter of SBI's weekly variance.

CHAPTER II

Beta & alpha
— *the market model*

THE MODEL

Excess return on excess return

Regress the stock's *excess* return on the market's *excess* return. The slope is β , the intercept is α , the leftover is ε .

MARKET MODEL (CAPM REGRESSION)

$$r_{i,t} - r_f = \alpha_i + \beta_i (r_{m,t} - r_f) + \varepsilon_{i,t}$$

β = systematic risk · α = Jensen's alpha · ε = idiosyncratic residual.

*Excess returns subtract the safe rate — only the part of the return that takes **real risk** remains.*

TWO VIEWS OF β

$$\beta = \text{Cov}(\text{stock}, \text{market}) \div \text{Var}(\text{market})$$

β has both a **statistical** definition and a **regression** definition.
They are the same number.

SLOPE OF THE MARKET-MODEL REGRESSION

$$\beta_i = \frac{\text{Cov}(r_i^{\text{ex}}, r_m^{\text{ex}})}{\text{Var}(r_m^{\text{ex}})} = \rho \cdot \frac{\sigma_i}{\sigma_m}$$

$\beta > 1$ aggressive · $\beta < 1$ defensive · $\beta \approx 1$ market-like.

β is **invariant** to a constant shift — using raw or excess returns gives the same β . But α is not; that is why CAPM is written in excess-return space.

EXCESS RETURNS

From annual to weekly — by compounding

The risk-free rate is quoted yearly. Convert it to the weekly horizon by **compounding**, not by simple division.

WEEKLY RISK-FREE RATE (COMPOUNDED)

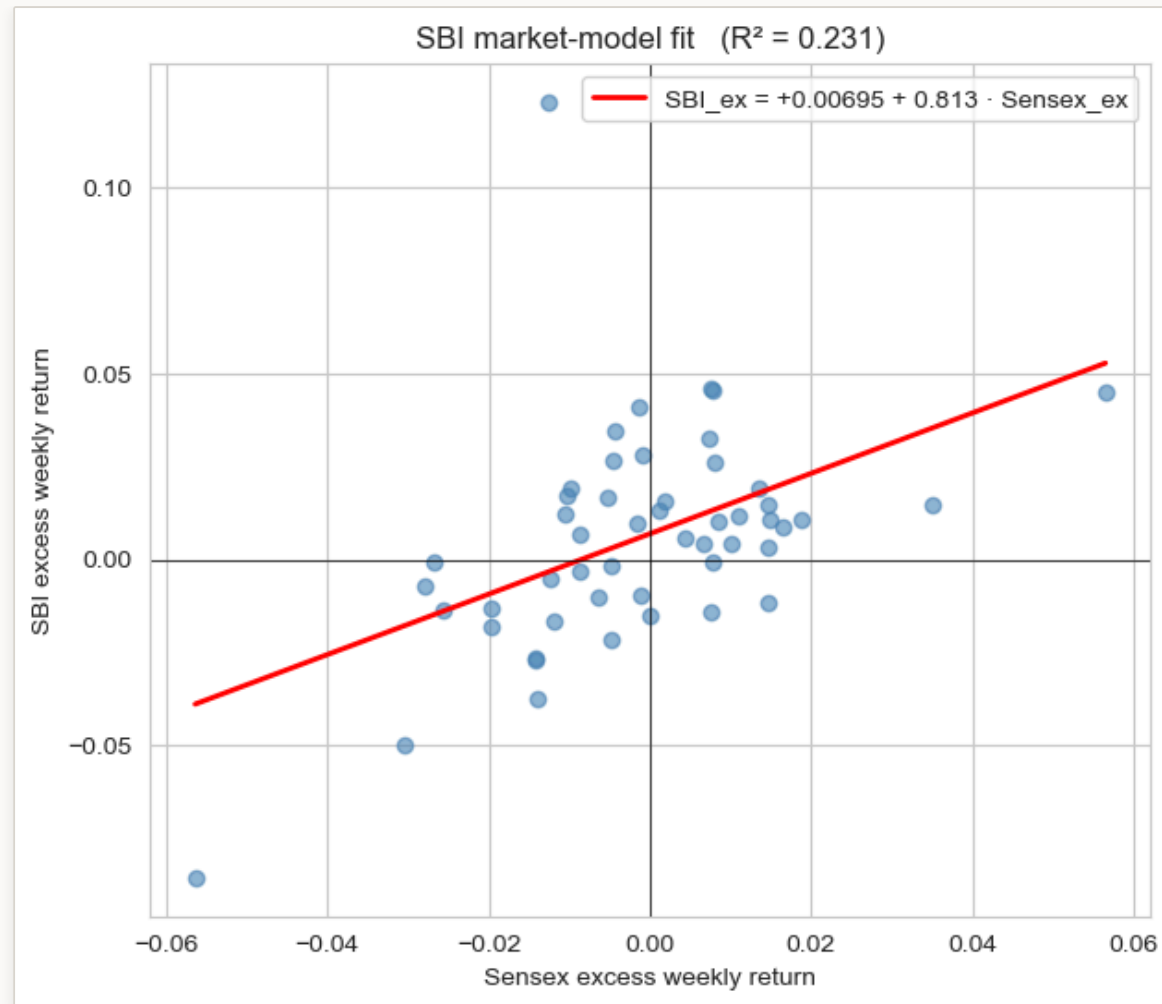
$$r_f^{\text{weekly}} = \left(1 + r_f^{\text{annual}}\right)^{1/52} - 1$$

Annual 6.5% → weekly 0.001212 ≈ 0.1212% per week. Subtract from every weekly return to form excess returns.

Annualise α the same way: $(1 + \alpha_{\text{weekly}})^{52} - 1$. Anything else is internally inconsistent.

SBI's market-model regression

- ▶ Slope (β) = **0.813** — mildly defensive
- ▶ Intercept (α) = **+0.70%** per week
- ▶ $R^2 = 0.231$ — market explains a quarter



RESULTS · OLS VIA STATSMODELS

SBI vs Yes Bank — head to head

	A (WEEKLY)	A (ANNUALISED)	B	SE(B)	R ²
SBI	+0.70%	+43.4%	0.813	0.212	0.231
Yes Bank	+0.35%	+20.2%	1.137	0.297	0.230

SBI is **defensive** ($\beta < 1$) with a larger headline α . Yes Bank is **aggressive** ($\beta > 1$) with a smaller α . But headlines aren't tests — how big are the standard errors?

HYPOTHESIS TEST

Is α distinguishable from zero?

statsmodels prints the test for free in `.summary()`. $H_0: \alpha = 0$ vs $H_1: \alpha \neq 0$.

T-STATISTIC AND REFERENCE DISTRIBUTION

$$t = \frac{\hat{\alpha}}{\text{SE}(\hat{\alpha})} \sim t_{n-2} \quad \text{under } H_0$$

Two coefficients estimated — intercept and slope — so degrees of freedom are $n - 2 = 49$.

SBI: $t = +1.89$, $p = 0.065$ · **Yes Bank:** $t = +0.68$, $p = 0.498$. Neither rejects $\alpha = 0$ at $\alpha_{\text{level}} = 0.05$.

DECISION · $\alpha = 0$?

Headline α , statistical zero

	$\hat{\alpha}$ (WEEKLY)	SE($\hat{\alpha}$)	T	P	DECISION @ 0.05
SBI	+0.70%	0.37%	+1.89	0.065	fail to reject
Yes Bank	+0.35%	0.52%	+0.68	0.498	fail to reject

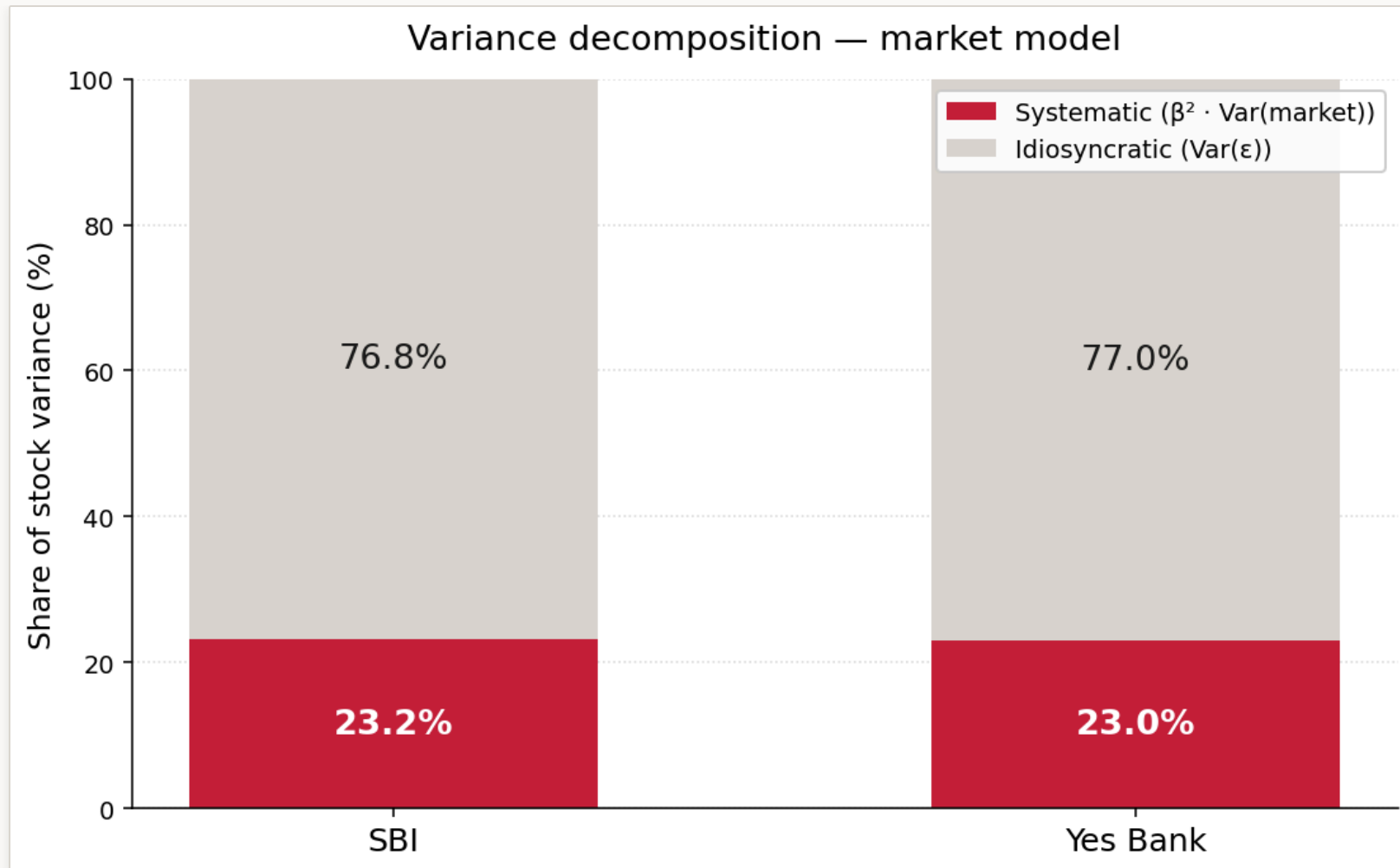
*With ~50 weekly observations, **Jensen's α is rarely significant**. The standard error is comparable to the point estimate — the data cannot tell apparent skill from noise.*

Systematic vs idiosyncratic variance

▶ Market explains ~23% for either stock

▶ **Three quarters** is stock-specific noise

▶ Diversification kills the grey, not the crimson



CLOSING

Five things to remember

- **Correlation** is symmetric, unit-free, and bounded — *regression* picks a direction and gives units.
- $R^2 = \rho^2$ — the same number, two different questions.
- $\beta = \text{Cov}(\text{stock}, \text{market}) / \text{Var}(\text{market})$ — the slope of the market-model regression.
- α is risk-adjusted return — and is rarely significant in a single year of weekly data.
- Convert annual rates to other horizons by **compounding**, never by simple division.

β tells you what you cannot diversify away.

— α tells you what is left for skill, if you can measure it.